

Solution: Given, $u, v \in \mathbb{R}^n$ and $u \neq v$. $T(u) = T(v)$.

By Theorem 10 from Chapter 1, A is the standard matrix of T of size $n \times n$. $\therefore Au = Av \Rightarrow Au + (-Av) = Av + (-Av)$

$$\Rightarrow Au + (-Av) = 0 \text{ (Algebraic property iv)}$$

$$\Rightarrow Au + (-1 \cdot Av) = 0 \text{ (Algebraic property iv)}$$

$$\Rightarrow Au + A \cdot (-1 \cdot v) = 0 \text{ (Theorem 5(b) from Chapter 1)}$$

$$\Rightarrow A(u + (-1) \cdot v) = 0 \text{ (Theorem 5(a) from Chapter 1)}$$

$$\Rightarrow A(u - v) = 0 \text{ (By defn.)}$$

$$\Rightarrow T(u - v) = 0 \text{ (Theorem 10 from Chapter 1)}$$

$\therefore T(x) = 0$ has a nontrivial soln.

By Theorem 11 from Chapter 1, T is not one-to-one.

$\therefore x \mapsto Ax$ is not one-to-one.

By the Invertible Matrix Theorem from Chapter 2 and by the equivalence of the statements (f) and (i), T does not map $\mathbb{R}^n$ onto $\mathbb{R}^n$. $\square$

39) Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be an invertible linear transformation, and let S and U be functions from $\mathbb{R}^n$ into $\mathbb{R}^n$ such that $S(T(x)) = x$ and $U(T(x)) = x$ for all x in $\mathbb{R}^n$. Show that $U(v) = S(v)$ for all v in $\mathbb{R}^n$. This will show that T has a unique inverse, as asserted in Theorem 9. [Given any v in $\mathbb{R}^n$, we can write $v = T(x)$ for some x . Why? Compute $S(v)$ and $U(v)$.]

Solution: By Theorem 10 from Chapter 1, A is the standard matrix of T of size $n \times n$, and $T(x) = Ax$ for all $x \in \mathbb{R}^n$. By Theorem 9, since T is invertible, so A must also be invertible.

• By Theorem 8 and the equivalence of the statements (a) and (i), we know that the linear transformation $x \mapsto Ax$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
• By Defn, for all v in $\mathbb{R}^n$, there exists x in $\mathbb{R}^n$ s.t. $v = T(x)$.

$$\therefore v = Ax$$

$$\text{Given } S(T(x)) = x \Rightarrow S(v) = x \Rightarrow S(Ax) = x$$

$$\text{Given } U(T(x)) = x \Rightarrow U(v) = x \Rightarrow U(Ax) = x$$

$$\therefore S(v) = U(v) \quad \forall v \in \mathbb{R}^n$$

$$\therefore T \text{ has a unique inverse} \quad \square$$